Jianfeng Zhang

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Education

- **Wuhan University** Wuhan, China
  Bachelor of Science in Applied Mathematics; GPA: 3.64; Ranking: 3/42 *Sep. 2014 – Jun. 2018*

Research Interests

- My research interests include 2D/3D Vision, Video Analysis, Deep Learning and their applications. Recently, I am focusing on semantic segmentation on 2D/3D images, the combination of semantic segmentation and stereo matching, and video classification.

Publications

- **Jianfeng Zhang**, Yan Zhu, Xiaoping Zhang, Ming Ye, Jinzhong Yang. *Developing a Long Short-Term Memory (LSTM) based Model for Predicting Water Table Depth in Agricultural Areas, Journal of Hydrology* (**JH, IF:3.73**), 2018.

- **Jianfeng Zhang**, Liezhuo Zhang, Yuankai Teng, Xiaoping Zhang, Song Wang, Lili Ju. *Interactive Binary Image Segmentation with Edge Preservation*, In submission to **AAAI**, arXiv preprint, 2018.

Projects

- **Interactive Binary Image Segmentation with Edge Preservation** Wuhan University, China
  Research Assistant *Mar. 2018 – Present*
  - Built an interactive image segmentation algorithm within Markov Random Field framework. We employed geodesic distance to compute unary term to separate regions with similar color appearance that belong to different labels. The bilateral affinity term was used as the pairwise term, which produced segmentation results with good boundary sensitivity.
  - The overall optimization problem was split into two subproblems through alternating direction strategy, which could be effectively solved via steepest gradient descent and fast bilateral solver.
  - This algorithm could reach 0.834 mean IoU in VGG interactive image segmentation dataset. Our research was preprinted at arXiv and I served as first author. [paper]
- **Long Short-Term Memory (LSTM) for Water Table Depth Prediction** Wuhan University, China
  Research Assistant *Apr. 2017 – May. 2018*
  - Developed a LSTM-based model to forecast water table depth in Hetao Irrigation District. The final best model achieved 0.93 R^2 score on prediction tasks.
  - Built a GUI to construct a decision system based on this model, to aid engineers and decision-makers to plan and manage groundwater resources in Hetao Irrigation District.
  - Our research was published in Journal of Hydrology, and I served as first author.[paper] [code]
- **Video Object Segmentation Framework** Wuhan University, China
  Research Assistant *Feb. 2018 – Mar. 2018*
  - We built a video object segmentation framework. In this framework, we used MobileNet to extract each frame's feature and utilized a simple algorithm to determine if a certain frame is key frame. Based on this method, we could extract expected number of key frames from a given video.
  - We utilized YOLO and Deep Grabcut to detect center object and obtain object binary mask, respectively.
  - Finally, we used key frame masks to finetune segmentation network before segment the whole video. Only 200-3000 iters of online finetune could ensure very good video object segmentation results.
  - This framework is currently applied to extract products in E-commerce videos and people in videos from Douyin, Kuaishou and YouTube.
- **DeeCamp: Human Activity Recognition** *Jul. 2018 – Aug. 2018*
  - I was admitted to a summer camp called DeeCamp hosted by Sinovation Ventures (acceptance rate: **4.2%**). Our team's task was human activity recognition based on video data.
  - I served as co-leader and responsible for implementing different video activity recognition models and building final demo.

- Our final demo was an online web game called "You Perform, I Guess!". This demo received **Excellent Demo Award** in DeeCamp.[demo] [code]

- **Clothing Fashion Style Recognition** *Jun. 2018 – Jul. 2018*
  - We built our fashion style recognition model based on ResNet, DenseNet and VGG. We also used Deeplab V3+ to extract people or clothes and remove background, which could help to denoise. In addition, we applied model ensemble method to improve our results. Finally, our method achieved 0.610 F2 score and ranked **9/213** on JD AI Fashion-Challenge.

- **End to End Network Image Text Detection and Recognition** *Mar. 2018 – May. 2018*
  - We Built a faster-rcnn like model for end to end network image text detection and recognition. Our method achieved 0.634 accuracy and ranked **13/1488** on Tianchi Challenge.

Work Experience

- **Farsee2 Inc.** Wuhan, China
  Research Intern with Prof. Xiaoping Zhang *Nov. 2017 – Present*
  - Reviewed several semantic and instance segmentation models and reimplemented Deeplab V3+ in PyTorch (this code receives more than **80 stars** in GitHub). We applied Deeplab V3+ to extract the expected object from multi-view images for stereo matching, in order to get better 3D reconstruction results.
  - Currently, we develop some new methods based on Deep Learning models for RGB-D semantic segmentation. In addition, we seek to build a model that can perform joint learning for semantic segmentation and disparity estimation, in order to improve both the semantic segmentation and stereo matching results.
- **SinoCLC Inc.** Beijing, China
  Data Mining Intern *Jul. 2017 – Sep. 2017*
  - Developed a web crawler to collect daily text data from Weibo and News Websites using Python with Scrapy.
  - Built a text classification system using CNN, RNN and Word Embeddings to classify articles and extract key words.

Awards & Honors

- **Excellent Demo Award, DeeCamp** *2018*
- **Best Bachelor Thesis Award (1/180+, School of Mathematics and Stastics)** *2018*
- **Award for Merit Student of Wuhan University** *2015-2017*
- **Honorable Mention, MCM, COMAP** *2015*

Teaching Experience

- **C Programming Language** *Spring 2017*
- **Data Structures and Algorithms in Python** *Fall 2017*

Programming Skills

- **Languages: Python, MATLAB, C/C++**
- **Deep Learning Tools: PyTorch, Tensorflow, Keras, Linux Shell, LaTeX**